

Adding Integers: Same Signs

Adding Integers with the Same Sign

When adding numbers with the same sign, there is no battle. Instead, the numbers **join forces** to create a larger asset or a deeper debt.

Example 1. **Joining Assets**

If you have \$400 in your savings account (+400) and you win \$150 (+150), you have a net balance of **\$550**.

$$400 + 150 = 550$$

Example 2. **Joining Debts**

If you owe your brother \$10 (−10) and then borrow another \$25 (−25), your net balance is a **debt** of **\$35**. So, you are falling deeper into debt.

$$-10 + (-25) = -35$$

We can now write a formal mathematical rule for adding integers with the same sign. Once again, we will use absolute value.

Procedure 1 (Steps for Adding Integers with the Same Signs). (a) Find the **absolute value** or **strength** of each number.

(b) **Add** the absolute values.

(c) Write the answer with the common sign.

Example 3. Use the three-step process to find the sum $-50 + (-30)$.

Explanation. Step 1: Find the Absolute Values (Strengths):

Find the strength of each number.

$$|-50| = 50 \quad \text{and} \quad |-30| = 30$$

Step 2: Add the Absolute Values (Join Forces):

Since the signs are the same, we add their absolute values.

$$50 + 30 = 80$$

Adding Integers: Same Signs

Step 3: Determine the Sign and Write the Answer:

Since both numbers are negative, the answer is negative.

$$-50 + (-30) = -80$$

Problem 1 Apply the steps to find the sum $-15 + (-40)$.

Step 1: Find the Absolute Values (Strengths):

$$|-15| = 15 \text{ and } |-40| = 40$$

Step 2: Add the Absolute Values (Join Forces):

$$15 + 40 = 55$$

Step 3: Determine the Sign and Write the Answer:

$$-15 + (-40) = -55$$

Problem 2 You add a \$20 credit (+20) to your gaming account and later add another \$15 credit (+15). Find your net balance.

$$20 + 15 = 35$$

Problem 3 An activist is arrested at a peaceful protest. A commercial bail bondsman charges a non-refundable \$500 premium (−500) to secure their release. Because the activist cannot pay upfront, the bondsman adds a \$150 financing fee (−150) to the debt. Find the activist's net balance.

$$-500 + (-150) = -650$$

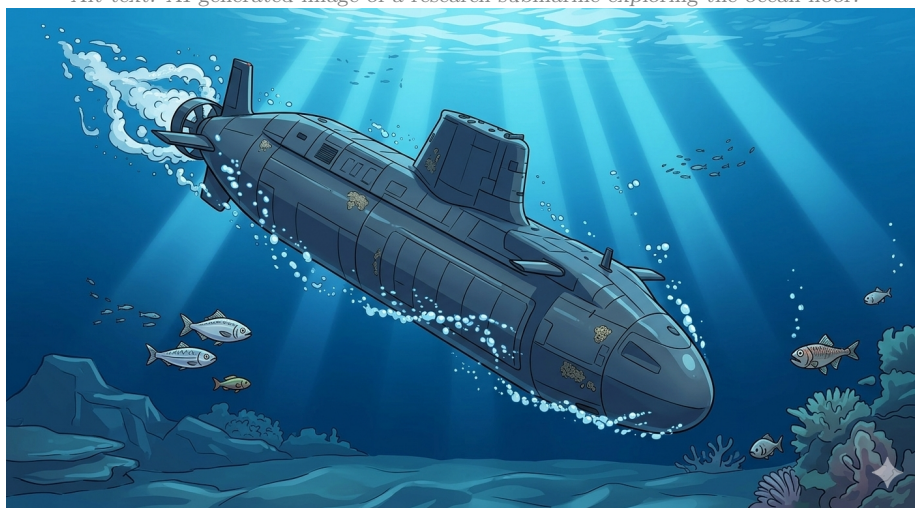
Adding Integers: Same Signs

Problem 4 During a football game, the home team's running back gained 6 yards on the first play of the drive. On the very next play, he charged forward an additional 2 yards. Find the total number of yards gained by the running back over these two plays.

$$6 + 2 = 8 \text{ yards}$$

Problem 5 A research submarine is exploring the ocean floor. It starts recording at a depth of 5 meters below sea level. The pilot then navigates the submarine down another 8 meters. Find the submarine's final depth relative to sea level.

Alt text: AI generated image of a research submarine exploring the ocean floor.



$$-5 + (-8) = -13 \text{ meters}$$

Problem 6 On a cold winter night in Cleveland, the starting temperature at midnight was -1°F . Over the next three hours, another cold front moved through the area and the temperature dropped by 5°F . Find the temperature at 3am.

Adding Integers: Same Signs

Alt text: AI generated image of a cold winter night in Cleveland.



$$-1 + (-5) = -6^{\circ}F$$